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Dissertation Brief

Duty Cycle-Aware Battery Degradation Analytics for Electric Vehicle Fleet Applications

Overview

This dissertation aims to develop a duty cycle-aware battery intelligence and preventive maintenance framework for electric vehicle (EV) fleets using explainable machine learning techniques. As EV adoption accelerates globally, understanding how real-world operational patterns influence battery degradation has become a major challenge for fleet operators, OEMs, and battery management system developers.

Unlike conventional studies that focus only on battery degradation prediction using laboratory datasets, this research investigates the relationship between dynamic vehicle usage patterns, powertrain load behaviour, and battery ageing characteristics. The study integrates open-source battery degradation datasets with representative EV drive cycle profiles to model how operational stress factors contribute to degradation progression.

The research will develop a machine learning framework capable of:

- identifying battery health states,
- predicting degradation trajectories,
- correlating duty cycle severity with ageing behaviour,
- and supporting fleet-level preventive maintenance decisions.

The work is fully simulation and data-driven, implemented using Python and MATLAB, with no physical experimentation or human participants involved.

Objectives

1. Review existing literature on EV battery degradation, fleet battery intelligence, drive cycle analysis, and predictive maintenance frameworks.
2. Analyse and compare open-source battery degradation datasets across different chemistries, capacities, and cycling conditions.
3. Develop operational stress indicators derived from representative EV drive cycles, including WLTP, UDDS, FTP-75, HWFET, and urban driving profiles.
4. Correlate dynamic current demand profiles and operational duty cycles with battery degradation behaviour and capacity fade characteristics.

5. Perform feature engineering to derive degradation-sensitive indicators such as: State of Health (SoH), internal resistance growth, coulombic efficiency, degradation rate, equivalent full cycles, current stress factors, and thermal exposure indicators.
6. Benchmark multiple machine learning models including Random Forest, XGBoost, Gradient Boosting, Support Vector Machine (SVM), and LSTM for: battery fault classification, degradation prediction, and Remaining Useful Life (RUL) estimation.
7. Develop an explainable AI-based battery degradation framework using feature importance and SHAP-based interpretability analysis.
8. Construct a fleet-level preventive maintenance and battery severity ranking framework based on operational stress and predicted degradation.
9. Relate findings to UK EV policy initiatives including the Zero Emission Vehicle Mandate, battery sustainability targets, and the Faraday Battery Challenge.

Datasets

The research will combine battery degradation datasets with representative EV operational drive cycle profiles to better reflect real-world fleet battery usage conditions.

Fleet datasets

1. Geotab EV Fleet Data
2. Open Energy Data Initiative Datasets
3. NREL Fleet & Transportation Datasets

Battery Degradation Datasets

1. NASA Prognostics Battery Dataset

Provides lithium-ion charge/discharge cycling data, impedance measurements, and capacity degradation trajectories under controlled ageing conditions.

2. Center for Advanced Life Cycle Engineering Battery Dataset

Contains battery ageing and impedance datasets across multiple chemistries and operational conditions.

3. University of Oxford Battery Degradation Dataset

Includes high-fidelity degradation experiments and dynamic cycling behaviour suitable for machine learning-based ageing analysis.

Operational Drive Cycle Profiles

Representative EV driving cycles will be incorporated to derive dynamic current demand and operational stress indicators:

- WLTP
- UDDS
- FTP-75
- HWFET
- NEDC
- Urban stop-go drive cycles

These profiles will be used to model realistic powertrain-induced battery stress conditions and correlate operational behaviour with degradation patterns.

Novelty and Research Contribution

The primary contribution of this dissertation is the integration of operational drive cycle behaviour with battery degradation modelling for EV fleet applications.

Unlike conventional battery health prediction studies that rely solely on laboratory cycling data, this research introduces:

- duty cycle-aware degradation analysis,
- operational stress correlation,
- explainable machine learning interpretation,
- and fleet-level battery severity indexing.

The dissertation proposes a Fleet Severity Index (FSI) framework to quantify the influence of:

- dynamic current demand,
- charging behaviour,
- cycling stress,
- and operational drive conditions
on battery degradation progression.

This approach aims to bridge the gap between laboratory-based battery ageing studies and real-world EV fleet operational behaviour.

Methodology

The dissertation will follow a multi-stage analytical framework:

1. Data Acquisition and Preprocessing

- Collection and harmonisation of battery degradation datasets.
- Cleaning, normalisation, interpolation, and cycle segmentation.
- Synchronisation of operational drive cycle profiles with battery ageing characteristics.

2. Drive Cycle and Operational Stress Analysis

Representative EV drive cycles will be analysed to derive:

- current demand profiles,
- acceleration/deceleration behaviour,
- regenerative braking intensity,
- powertrain load fluctuations,
- and duty cycle severity indicators.

3. Feature Engineering

Battery degradation-sensitive features will be extracted, including:

- SoH,
- capacity fade,
- internal resistance growth,
- coulombic efficiency,
- equivalent full cycles,
- RMS current stress,

- C-rate variability,
- and operational stress factors.

4. Machine Learning Modelling

The following models will be benchmarked:

- Random Forest,
- XGBoost,
- Gradient Boosting,
- Support Vector Machine,
- and LSTM networks.

The models will be evaluated for:

- battery health classification,
- degradation prediction,
- and Remaining Useful Life estimation.

5. Explainability and Interpretability

SHAP analysis and feature importance methods will be used to identify the dominant operational factors contributing to battery degradation.

6. Fleet-Level Maintenance Framework

A preventive maintenance decision framework will be developed to:

- rank fleet battery severity,
- identify high-risk operational patterns,
- and support predictive maintenance scheduling.

7. Evaluation Metrics

Models will be evaluated using:

- Accuracy,
- Precision,
- Recall,
- F1-score,
- RMSE,
- MAE,
- and cross-validation techniques.

Note on Powertrain-Integrated Approach

This dissertation adopts a powertrain-aware battery intelligence approach by incorporating representative EV drive cycle profiles and operational load characteristics into battery degradation analysis.

Although direct fleet-level powertrain telemetry data is commercially restricted and not publicly accessible, operational current demand behaviour can be realistically approximated using standardised

and real-world driving cycles. These drive cycles provide representative acceleration, deceleration, regenerative braking, and load fluctuation patterns that directly influence battery stress and ageing behaviour.

The research therefore integrates:

- drive cycle-derived dynamic load characteristics,
- battery ageing datasets,
- and machine learning-based degradation analysis to establish a practical approximation of real-world fleet battery operational behaviour.

This approach enables the dissertation to bridge laboratory degradation datasets with vehicle-level operational stress conditions while remaining fully simulation and data-driven.

Completion Plan

The dissertation will be completed across six phases between May and early September 2026:

Phase	Timeline	Key Activities
Setup & Literature Review	May 2026	Supervisor confirmation, EG7030 registration, literature review, dataset download, Python/MATLAB environment setup.
Data Preparation	Early June 2026	Exploratory data analysis, feature engineering, data cleaning and normalisation across all three datasets.
Model Development	Mid June – July 2026	Fault classification (Random Forest, XGBoost) and degradation prediction (LSTM). Tuning and cross-validation.
Framework Design	Late July 2026	Preventive maintenance decision framework built from model outputs. Powertrain integration documented at conceptual level.
Results & Analysis	Early August 2026	Results interpreted, visualisations prepared, discussion and limitations drafted.
Writing & Submission	August – Early Sept 2026	Full dissertation written (~12,500 words), supervisor review completed, submitted via UEL portal.

I am happy to discuss any aspect of this proposal further. Thank you for your time and consideration.